# Neurological Foundations for Listening and Talking

**DEVELOPMENT OF AUDITORY FUNCTION**

**FOUNDATIONS OF AUDITION**

**ACOUSTICS OF SPEECH**

# Learning outcomes



- Understand development of auditory function, and how it is disrupted due to untreated hearing loss (neuroplasticity, cross modal reorganization)
- Understand the importance of listening experience for children with hearing loss
- Understand the frequency, intensity, and temporal parameters of speech acoustics
- Recognize the developmental milestones from the auditory development scale

# Development of auditory function

- What is auditory function
- Foundations of Audition
  - Brain development/neuroplasticity, rationale for early access
  - Cross modal reorganization
  - Hearing vs listening
  - Integrating listening into whole child
- Levels of auditory function/listening hierarchies
  - Following normal development
  - Detection, discrimination, identification, comprehension

# Auditory Function



- The ability of the brain to use sound meaningfully
- Auditory sound waves are converted into patterns of neural activity which are integrated with other sensory input - which from a survival standpoint may lead us to move (alerting) and at more sophisticated levels allow us to respond to music and to communicate with one another.
- Hearing is the primary sense
  - "…listening (audition) is the primary modality for learning spoken language…" Cole, E. B., Flexer, C. A. (2007). Auditory "Work". In Children with hearing loss: developing listening and talking. San Diego: Plural Pub., P. 3.

# The foundation to Auditory learning is access to sound



- Ongoing **audiologic management** and listening with hearing devices

# The task of auditory function



| Pick up | Change | Send | Process |
| --- | --- | --- | --- |
| Pick up the signal | Change it into neural responses | Send the signal to the correct areas of the brain | Process the signal meaningfully and respond as needed |

# Auditory Function and neuroplasticity

Carol Flexor, “We hear with the brain”

Where it starts: ears are “the doorway to the brain”

The pathway to the correct part of the brain has to be forged.

# Typical Infants

- Cochlea is formed and functional by the **20th** week of gestation
- Listening is occurring in utero before birth
- Infants are born with **20 weeks** of listening experience
- Newborns with hearing loss may have already missed 20 weeks of listening

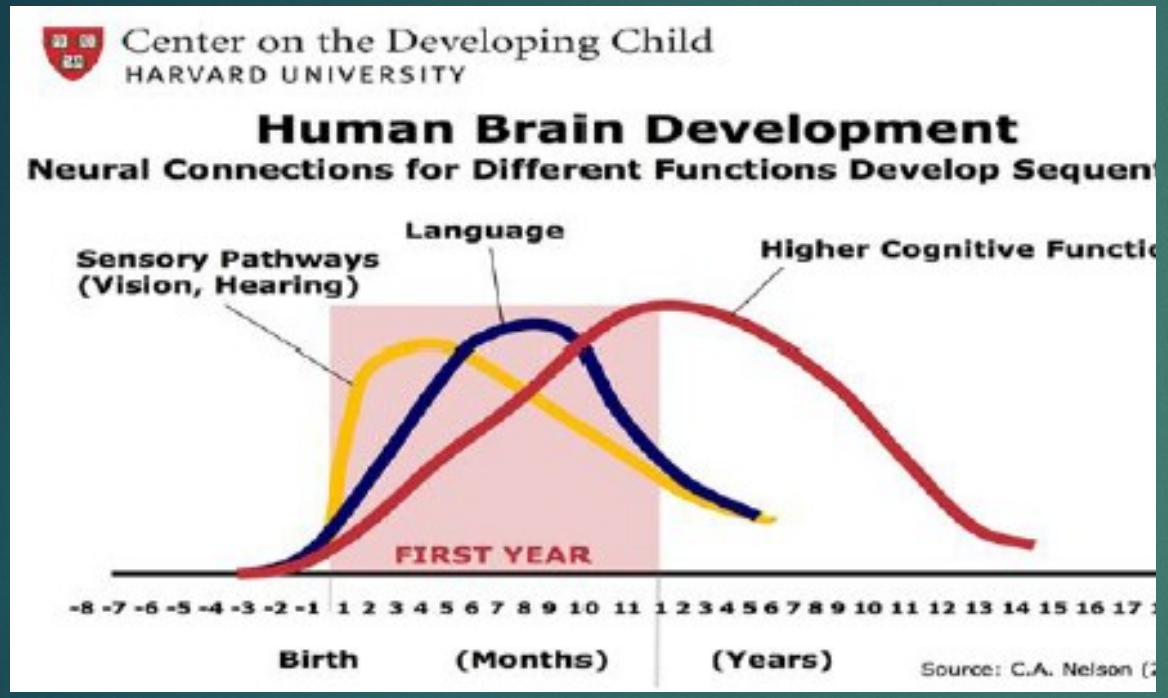
# It's about the brain



- **Over 1 million** neural connections form per second in the first few years of life
- Newborns Distinguish Speech
  - Children as young as four days old can distinguish the vowel sounds of the language in their natural environment from those of a foreign language

# Functional reorganization

By end of 1st year there is a **functional reorganization** of the brain to distinguish phonemes specific to language(s) heard daily
This neural  reorganization improves and tunes the phonetic categories required for the infant's language and attenuates those phonemic distinctions not required for their language
**Auditory experience** is critical for development of speech and language




Normally, in the first year of life, the brain develops the pathways for **vision**, hearing, and **language**.

# Hearing loss is an emergency

- If in normal development the foundational pathways for vision, hearing, and language are set by 12 months, what does this mean for children who first get good access to sound with a cochlear implant at 12-13 months of age?
- Babies with hearing loss become delayed.
- Hearing Loss is a **neurobiological emergency**

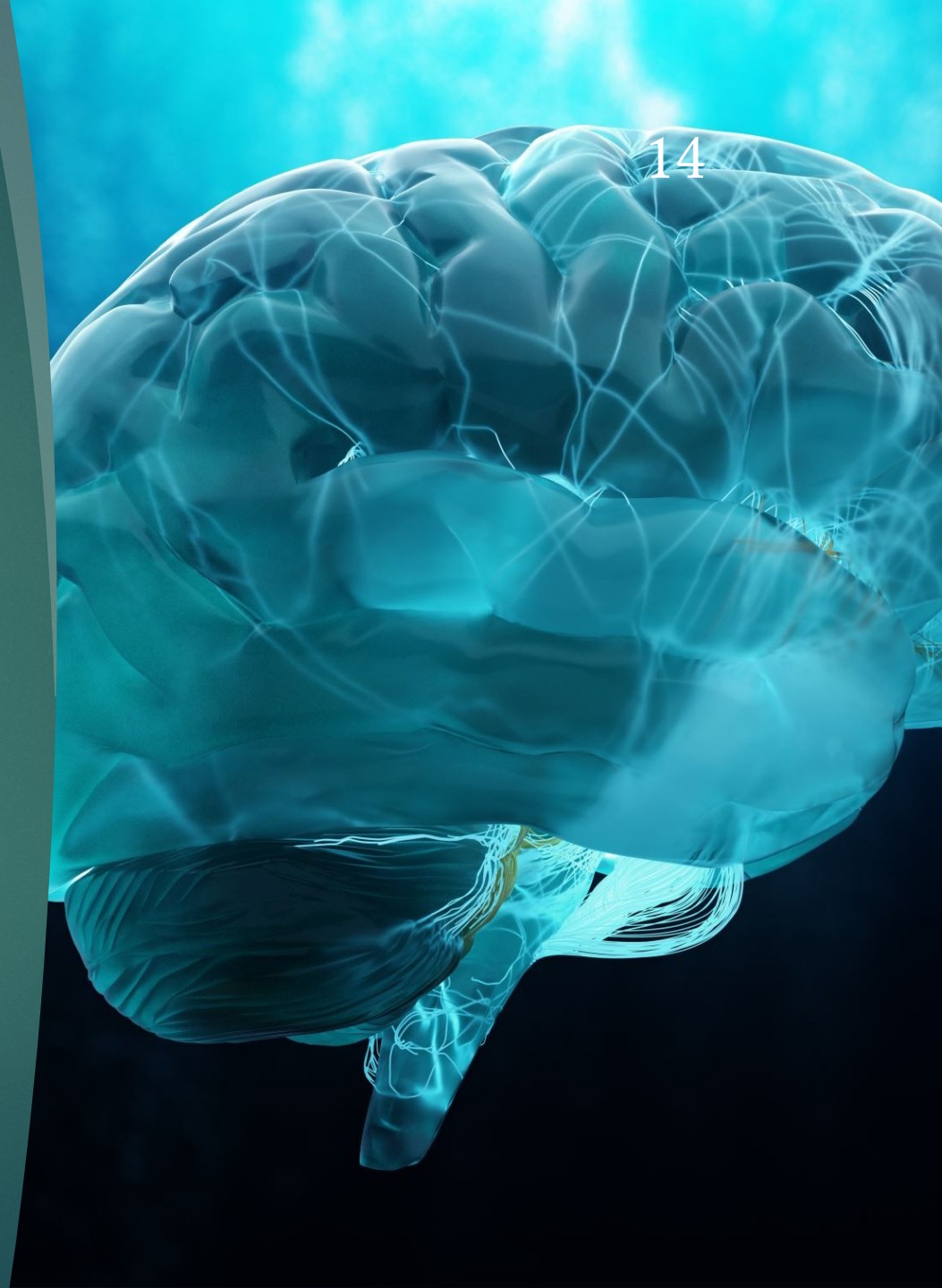


# You Speak How You Hear

- Young children will speak how their parents speak
- If your auditory input is poor, your speech will be too
  - **Muddy in, muddy out**
- Parents and Caregivers are the architects of early brain development.
- Audiologists and SLP's must teach, coach, cheer, listen, and facilitate full-time device use.
  - Think: **Eyes open, technology on**

# Neuroplasticity

- Brain's ability to grow, develop and alter its structure, as a function of external stimulation
- We can now stimulate auditory brain centers during times of maximum neuroplasticity – **the first 3½ years of life (greater)**
- Auditory brain access is available to infants with profound losses – due to CI
- Early amplification or implantation stimulates a brain that is in the initial process of organizing itself and is therefore more receptive to auditory information, resulting in greater auditory capacity

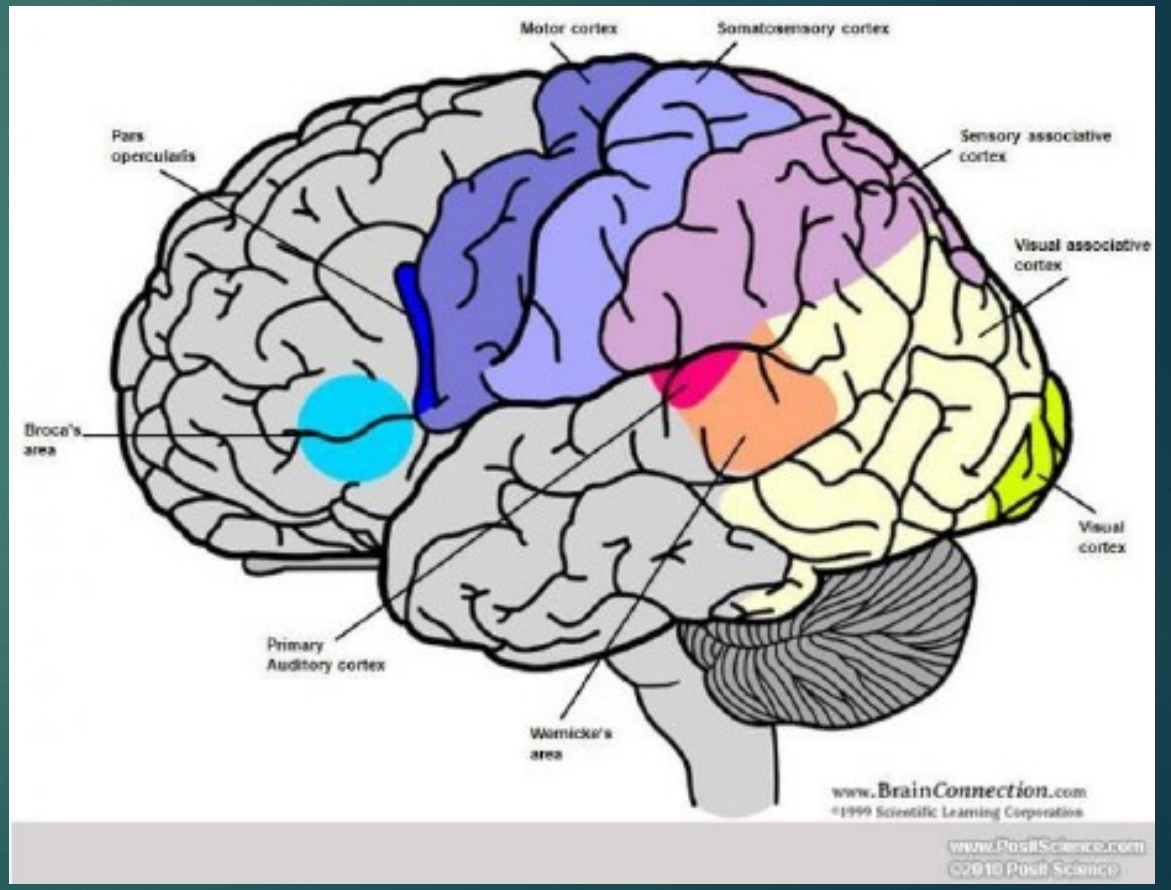


# Neuroplasticity

- Neurons can be thought of as the “workers awaiting a job assignments” in the brain
- The Neuroplasticity (ability of these “workers” to learn a new job) is greatest in the first 3 ½ years of life.
- Without auditory stimulation, to do their job, these workers forget how to do their hearing job.
- **Use it or lose it**

# Cross-Modal Reorganization

- "Use it or lose it"
- The primary auditory cortex is housed in **the Heschel's Gyrus** (dark pink area).
- If auditory stimulation is not received the brain reallocates use of this area to **vision and touch** (*Karns, Dow, and Neville(2012)*



# Developmental Synchrony



- Developmental Synchrony: **skills developing/mastered at age-appropriate times, biologically.**
- Child will be operating under a developmental paradigm not a remedial paradigm
- Delayed auditory development leads to delayed language, which necessitates a **remedial**, rather than **developmental paradigm**

# Hearing versus Listening



- Hearing is when the ear perceives sound.  With typical hearing, it just happens.
  - **Access to Sound**

- Listening is a conscious effort that requires concentration.  Listening is required to understand what someone is saying.
  - **Attention to Sound**

- Hearing ≠ Listening
  - Listening is a conscious activity.
- Young listeners will benefit from a cue to “listen”

Listening experience

↓

**Phonetic** categories

↓

Phonological processes

↓

**Lexical-semantic** use

↓

reading and higher order language use

# How much listening experience is necessary for adequate language development?

- About **20,000** hours in the first **5** years of life to create the neural infrastructure for literacy
- As a result, children receiving CI early may benefit more from the greater plasticity of the auditory pathways

# Listening hours per day



Babies with typical hearing

- 10 hours of listening per day
- 10 hours a day = 3, 650 hours of listening time per year.

Toddlers/preschoolers  with typical hearing

- 12 hours of listening per day
- 12 hours a day = 4,380 hours of listening per year

Babies with part time HA/CI use

- 4 hours of use per day
- It will take 6 years of listening time to equal the listening time of the 1 year old with typical hearing

Toddlers/preschoolers with part-time HA/CI use

- With 3 hours of use per day, it will take 9 years of listening time to = just one year of listening experience gained by the child with full-time device use (Rossi, 2003).

# Creating the listening environment



- Parents and interventionists can focus on developing the child's listening skills, strategies and choices only after the audiologist channels acoustic information to the brain by fitting and programming technologies, not before
- Guide and coach parents to create environments that support listening for the acquisition of spoken language throughout the child's daily activities.
- Guide and coach parents to help their child integrate listening and spoken language into all aspects of the child's life.

# Concept of Listening Age



- Child with normal hearing listens to spoken language for at least **10-12** months before they begin producing single words
- And another **10-12** months before combining words
- Child with hearing loss will need similar amount of time
- The day child begins wearing technology all waking hours is **Day Zero** for listening age
- If child begins at 3 months – will have a listening age of 9 months at 1 yr birthday

# Factors that may advance or delay listening age:

- Age at which **technology** was acquired
- **Quality** of speech signal made available
- Amount of **time** each day child wears technology
- Skill of family and therapist in providing auditory language enrichment

# Audibility versus Intelligibility



- Audible: ability to **detect presence**
  - Audibility: “ I hear it”
- Intelligible: **discriminate** the word-sound distinctions of individual phonemes
  - Intelligibility: “I understand it”
- Speech might be audible, but not intelligible

# Development of Auditory Skills



- Our ability to communicate verbally depends on the **quality** of our auditory perception of the various segmental and suprasegmental aspects of speech

- Detection capabilities
  - The human auditory system has sophisticated perceptual capabilities, but is limited in terms of the signals it can process
  - Figure 4.2 Intensity limits vary as a function of frequency of the auditory stimulus

# Speech Acoustics

# Intensity Parameters of Speech:



- Average intensity of speech = approx. **45** dBHL
- **Distance** is obvious factor
- Variability exists in acoustical energy of individual speech sounds
- Most powerful phoneme (schwa)– weakest phoneme (th)
- Male speakers produce speech at about **3** dB greater than females
- Table 4.1

Frequency Parameters of Speech:

Overall speech spectrum: **50–10,000** Hz
Greatest amount of speech energy is found in frequencies below **1000** Hz
Figure 4.3
Figure 4.5 – combined intensity and frequency values associated with individual speech sounds

# Temporal Parameters of Speech:



- Duration of individual speech sounds ranges from **20–400** msec
- **Vowels** generally have longer duration and higher acoustic energy, low-mid freq range… as opposed to **consonants.**
- Individual phoneme duration does contribute to speech perception

# Transitional Cues



- Intensity, frequency and durational properties are altered significantly when phoneme is produced with other phonemes in conversation
- In conversational speech, the dynamic movement of the **articulators** in the production of adjacent phonemes produces **acoustical by-products**
- These are important for speech perception

# Summary:



- Vowel sounds have strong **low-mid** frequency energy
- Vowels: 250Hz-**500**Hz
- Vowels carry **90**% of energy of speech
- Consonant sounds are very weak, high-frequency sounds
- Consonants: **2000**Hz/4000Hz and above
- Consonants carry only **10**% of energy of speech, but **90% of the information needed** to perceive the differences among sounds
- Octave band at **2000**Hz carries greatest amount of speech information
- For speech to be perceived clearly, both vowels and consonants must be acoustically available to the brain

# Ling 10 sound test



- Allows a quick and easy way to verify that a child **detects** the vowel and consonant sounds of spoken language
- Sounds selected for this test were selected because they encompass the entire **speech spectrum**
  - Allows parents and teachers to know the child's **distance hearing**
- Frequent administration of this test can help parents and professionals monitor for:
  - Hearing aid malfunction
  - CI malfunction
  - Changes in child's hearing
  - Onset of middle ear pathology

# Advanced Bionics Ling 10 Handout on Moodle, Page 2

https://blog.medel.com/the-ling-six-sound-test-explained/

# Redundancy and Noise:



- Perception of speech involves more than just acoustics of speech or acoustic abilities of the listener
- Conversational speech is **highly redundant**
- Table 4.3
- Table 4.4 – Syntactic Constraints, Semantic Constraints, Situational Constraints
- Table 4.5 Sources of Noise in Oral Communication

# Normal Speech and Language Development

# Normal Development



- Based on the age by which ***most*** monolingual speaking children will accomplish identified milestones

- All items in a category are not typically mastered until reaching the **upper age limit** in the age ranges.

- Why important?
  - Not just so you know what abnormal development is…
  - To aid in counseling a family regarding progress

# Integrated Scales of Development



- Document on Moodle
- Integrated scale that outline typical stages of development in the areas of listening, receptive and expressive language, speech, cognition and social communication. They have been adapted from a number of sources:
  - Cottage Acquisition Scales for Listening, Language and Speech
  - Preschool Language Scale – 4 (PLS 4)
  - The Bzoch-League Receptive-Expressive Emergent Language Scale Second Edition
  - (REEL - 2)
  - The Early Learning Accomplishment Profile Kit (E-LAP)
  - The Learning Accomplishment Profile Revised Edition Kit (LAP-R)
  - The Rosetti Infant-Toddler Language Scale
  - St. Gabriel's Curriculum

# Vocabulary

- How is vocabulary learned?

- 18 months = 10 to **15** words
- 24 months = **40–50** words words
- 2 years = 150 words
- 3 years = 300 to 400 words
- 3 to 4 years = 600 to 1000 words
- 4 to 5 years = **1000–1600** words
- 5 to 6 years = 1500 to 2100 words

(Talking Child, LLC)

# Vocabulary: Food For Thought



- Average number of words per hour addressed to children by parents (Hart & Risley, 1999):
  - 2100 in a professional family
  - 1200 in a working-class family
  - 600 in a welfare family

- How does the vocabulary differ among families?
  - Professional and working-class families
    - More varied vocabulary, complex ideas, subtle guidance, and positive feedback

# Typical Speech Intelligibility



| Approximate Age | % Understood by Parents |
|---|---|
| 18 months | A typical child is 25% intelligible |
| 2 years | A typical child is 50-70% intelligible |
| 3 years | A typical child is 80% intelligible |
| 4 years | A typical child is 90% intelligible |

(Talking Child, LLC, 2003)

# “The Invisible Acoustic Filter Effect”



- Hearing loss – a silent handicap
- Hearing loss can be described as an invisible acoustic filter that “distorts, smears, or eliminates incoming sounds, especially sounds from a distance – even a short distance.”
- Hearing is not an ***isolated*** activity.  Everything else is built upon hearing
- Hearing loss itself is invisible and can be easily ignored or underestimated.
  - Only the **neg**. **effects** of hearing loss are visible.

# Developing Listening and Talking:



- Early identification and intervention
- Vigilant and ongoing audiologic management
- Immediate and consistent brain access of auditory info via technology
- Guidance from a professional who is ***highly qualified*** in the development of listening and speaking through techniques of parent coaching
- Following the professional's coaching with daily and ongoing formal and informal auditory, language, cognitive, and literacy enrichment.
- Integration and use of auditory strategies into all-day, everyday interactions with the child

# Speech Perception and Comprehension: Listening Hierarchies



- Speech perception implies understanding and comprehension
- Reception is the first step
- In it's most basic form, the perception of speech may be thought of as involving a number of important components:
  - Detection
  - Discrimination
  - Identification
  - Comprehension